

KESSLER

Software Requirements Specification (SRS)

Document ID: KSL-SRS-001 **Version:** 1.0 **Date:** 21 July 2026 **Status:** Baselined

Product: KESSLER — Open Orbital Conjunction Assessment & Risk Toolkit

1. Introduction

1.1 Purpose

This SRS defines the functional and non-functional requirements for **KESSLER**, an open-source conjunction assessment toolkit. KESSLER ingests public space-surveillance data (Conjunction Data Messages and General Perturbations orbital elements), computes and verifies collision probability (Pc), classifies orbital-collision risk, and presents results through a REST API and a 3D web dashboard. The document is the contract between the architecture and the implementation and is the baseline against which QA validates the release.

1.2 Scope

KESSLER addresses a verified gap in the space-safety ecosystem: the reference conjunction assessment tooling (NASA CARA) is MATLAB-based, and no lightweight open Python library exists for CDM parsing plus collision-probability computation, while the US TraCSS programme is beginning to distribute standardized CDMs to all satellite operators. KESSLER provides:

- A **backend service** (Python/FastAPI) with a clean separation of *logic* (astrodynamics and risk mathematics) and *service* (orchestration, data access, caching) layers.
- A **3D web dashboard** (React + Three.js) visualising conjunction geometry, orbits and risk.
- A **pipeline observability view** that renders the agent/data-flow design diagram (n8n-style) live in the product UI.

Out of scope for v1.0: manoeuvre planning, operator-grade CDMs with full covariance from Space-Track operator accounts, multi-tenant auth, and cloud deployment (AWS integration is planned; credentials/infrastructure will be provided in a later phase).

1.3 Definitions and Acronyms

Term	Definition
CDM	Conjunction Data Message (CCSDS 508.0-B-1) — standardized message describing a predicted close approach between two space objects.
TCA	Time of Closest Approach.
P _c	Probability of Collision for a conjunction event.
GP / OMM	General Perturbations orbital data / Orbit Mean-Elements Message (CCSDS 502.0-B-2).
TLE	Two-Line Element set — legacy GP data encoding, propagated with SGP4.
SGP4	Simplified General Perturbations propagator model #4.
RTN	Radial / Transverse (in-track) / Normal (cross-track) satellite-centred frame.
HBR	Hard-Body Radius — combined effective radius of the two objects.
Screening	Bulk search for close approaches between a protected asset and the catalogue.
TraCSS	US Office of Space Commerce Traffic Coordination System for Space.

1.4 References

- CCSDS 508.0-B-1 — Conjunction Data Message Blue Book.
- TraCSS-Spec-001 v2.1 — CDM Specification for TraCSS (Office of Space Commerce, 2026).
- Space-Track.org API documentation (cdm_public, gp classes; rate limits).
- CelesTrak GP data formats (gp.php, OMM JSON).
- Foster, J. L., Estes, H. S. (1992) — Parametric analysis of orbital debris collision probability.
- Chan, F. K. (2008) — Spacecraft Collision Probability (analytic series method).
- NASA CARA Analysis Tools (reference implementation, MATLAB).

2. Overall Description

2.1 Product Perspective

KESSLER is a self-contained web application plus reusable computation core. It consumes public external data services and requires no paid dependency. It runs fully on localhost for v1.0; the service layer is written so that storage and schedulers can later be re-pointed to AWS (S3/RDS/EventBridge) without touching the logic layer.

2.2 User Classes

User class	Description	Primary interest
Smallsat operator	Operates 1–20 satellites, no dedicated SSA team.	“Is anything coming close to my satellite this week, and how worried should I be?”
SSA analyst / researcher	Studies conjunction statistics and Pc methods.	Reproducible Pc computation, exportable data.
Developer	Integrates conjunction data into other systems.	Clean REST API, typed schemas, predictable errors.
Demo viewer	Evaluates the project (recruiter, community).	3D visualisation, pipeline diagram, polish.

2.3 Operating Environment

- Backend: Python ≥ 3.11 , FastAPI/uvicorn, Windows/Linux/macOS.
- Frontend: Node ≥ 18 , Vite build, evergreen browsers with WebGL 2.
- v1.0 target: localhost (backend :8000, UI dev server :5173 with API proxy).

2.4 Design Constraints

ID	Constraint
CON-1	Space-Track rate limits (<30 req/min, <300 req/hr; GP at most hourly) MUST be respected; architecture MUST be cache-first. Live per-user pass-through queries are forbidden.
CON-2	Redistribution: only basic SSA data (TLE/OMM, SATCAT, decay) may be re-served, and always with Space-Track citation. CDM-derived screens carry source attribution.
CON-3	The system MUST be fully functional with zero credentials (Demo Mode) using bundled sample data, because Space-Track accounts and AWS are provisioned later.
CON-4	No paid services in v1.0. Optional AI features degrade gracefully when no API key is configured.
CON-5	Repository layout: separate <code>ui/</code> and <code>backend/</code> top-level folders; backend separates <code>logic/</code> and <code>service/</code> packages.

3. Functional Requirements

3.1 Data Ingestion (FR-ING)

ID	Requirement	Priority
FR-ING-1	The system SHALL ingest GP/OMM orbital elements from CelesTrak (<code>gp.php</code> , JSON OMM format) without authentication.	Must
FR-ING-2	The system SHALL ingest public conjunction data from the Space-Track <code>cdm_public</code> class when credentials are configured (fields incl. CDM_ID, CREATED, TCA, MIN_RNG, PC, SAT_1_ID, SAT_1_NAME, SAT_2_ID, SAT_2_NAME, EMERGENCY_REPORTABLE).	Must
FR-ING-3	All external fetches SHALL be cached in a local store with per-source TTL (GP $\geq$ 1 h, CDM $\geq$ 30 min) and served from cache within TTL.	Must
FR-ING-4	In Demo Mode (no credentials) the system SHALL load bundled sample GP and CDM datasets and mark responses <code>data_mode: "demo"</code> .	Must
FR-ING-5	Ingestion failures SHALL degrade to last-known-good cache with a structured warning, never a hard UI failure.	Must

3.2 CDM Parsing (FR-CDM)

ID	Requirement	Priority
FR-CDM-1	The system SHALL parse CCSDS CDM messages in KVN format into a typed model (header, relative metadata, per-object metadata/OD/state/covariance sections).	Must
FR-CDM-2	The system SHALL parse Space-Track <code>cdm_public</code> JSON rows into the same internal model.	Must
FR-CDM-3	The parser SHALL validate mandatory fields (TCA, both object designators) and units, raising a typed <code>CdmParseError</code> listing all violations.	Must
FR-CDM-4	When RTN covariance and relative state are present the parser SHALL expose them for Pc computation; when absent it SHALL flag the event <code>covariance_available: false</code> .	Must

3.3 Collision Probability (FR-PC)

ID	Requirement	Priority
FR-PC-1	The system SHALL compute 2D Pc by the Foster method: project combined position covariance onto the encounter plane (perpendicular to relative velocity) and numerically integrate the bivariate Gaussian over the HBR disc.	Must
FR-PC-2	The system SHALL additionally compute Chan's analytic-series Pc as an independent cross-check; results SHALL agree within 5% for well-conditioned covariances, else the event is flagged <code>pc_method_divergence</code> .	Must
FR-PC-3	When covariance is unavailable the system SHALL compute a conservative maximum Pc (worst-case covariance orientation/scale, Alfano-style) and label the value <code>pc_type: "max"</code> .	Must
FR-PC-4	Default HBR SHALL be 20 m combined, overridable per request; object-type heuristics (payload/rocket body/debris) MAY refine the default.	Should
FR-PC-5	All Pc computations SHALL be pure functions in the logic layer (no I/O) and unit-tested against reference cases, including the closed-form isotropic-covariance solution.	Must

3.4 Screening (FR-SCR)

ID	Requirement	Priority
FR-SCR-1	Given a protected asset (NORAD ID) and a screening window (≤ 72 h), the system SHALL find close approaches against the loaded catalogue subset using SGP4 propagation.	Must
FR-SCR-2	Screening SHALL apply a perigee/apogee altitude-band pre-filter ($\pm$ screening threshold) before propagation to bound cost.	Must
FR-SCR-3	Coarse sampling (≤ 60 s step) SHALL be refined to TCA by bounded scalar minimisation; reported TCA accuracy ≤ 1 s.	Must
FR-SCR-4	Screening results SHALL include TCA, miss distance (total and RTN components), relative speed, and max-Pc estimate.	Must
FR-SCR-5	A screening run over 500 catalogue objects $\times$ 24 h SHALL complete in ≤ 30 s on a mid-range laptop.	Should

3.5 Risk Assessment (FR-RSK)

ID	Requirement	Priority
FR-RSK-1	Events SHALL be classified using operator-community thresholds: CRITICAL $P_c \geq 1e-4$; WARNING $1e-5 \leq P_c < 1e-4$; MONITOR $1e-7 \leq P_c < 1e-5$; NEGLIGIBLE $P_c < 1e-7$. Miss distance < 1 km at TCA raises the class one level if covariance is unavailable.	Must
FR-RSK-2	Each event SHALL receive an urgency score combining risk class, time-to-TCA and data age, defining dashboard sort order.	Must
FR-RSK-3	Each assessment SHALL include a deterministic recommended action string (e.g. "Request operator CDM refresh; re-screen at T-12 h").	Must
FR-RSK-4	(Optional) When an Anthropic API key is configured, the system SHALL generate an AI analyst briefing per event; without a key it SHALL fall back to the deterministic template. Failures fall back silently with a logged warning.	Could

3.6 API (FR-API)

ID	Requirement	Priority
FR-API-1	REST API under <code>/api/v1</code> : list conjunctions (sorted by urgency), event detail, satellite ground/orbit track, run screening, pipeline status, health.	Must
FR-API-2	All error responses SHALL use a single envelope: <code>{"error": {"code", "message", "detail", "request_id"}}</code> with correct HTTP status (400/404/422/502/503/500).	Must
FR-API-3	OpenAPI schema SHALL be served at <code>/docs</code> (FastAPI native).	Must

3.7 Visualisation (FR-UI)

ID	Requirement	Priority
FR-UI-1	The dashboard SHALL render an interactive 3D Earth (WebGL/Three.js) with the protected asset's orbit, secondary-object orbits for selected events, and conjunction markers at TCA geometry.	Must
FR-UI-2	An event list SHALL show risk class, Pc, miss distance, relative speed and countdown-to-TCA, sorted by urgency; selecting an event focuses the 3D view and opens the detail panel.	Must
FR-UI-3	A Pipeline page SHALL embed the n8n-style agent/data-flow diagram (the same artefact delivered in <code>documents/diagrams/</code>) with live stage status from <code>/api/v1/pipeline/status</code> .	Must
FR-UI-4	Visual design: dark space-grade theme, purposeful; no decorative pattern noise. All numbers displayed with units and reference frame.	Must
FR-UI-5	API failures SHALL surface as non-blocking toasts with retry; the 3D scene keeps last-good data.	Must

4. Non-Functional Requirements

ID	Category	Requirement
NFR-1	Reliability	Every service-layer entry point wraps external I/O in typed exception handling; no unhandled exception may cross the API boundary (global handler guarantees the error envelope).
NFR-2	Observability	Structured logging (timestamp, level, logger, request_id, message) to console and rotating file <code>backend/logs/kessler.log</code> (5 MB × 3). Each request gets a request_id propagated into responses.
NFR-3	Performance	Cached list endpoints p95 < 200 ms; single-event Pc computation < 50 ms; UI first meaningful paint < 3 s on localhost.
NFR-4	Security	Credentials only via environment variables (never committed); CORS restricted to localhost dev origins in v1.0; no PII stored.
NFR-5	Portability	No OS-specific paths; SQLite file store abstracted behind a cache interface so AWS (S3/RDS) can replace it without logic-layer changes.
NFR-6	Maintainability	Logic layer is framework-free pure Python; ≥ 85% unit-test coverage on <code>logic/</code> ; type hints throughout; UI strict TypeScript.
NFR-7	Compliance	Space-Track citation rendered in UI footer and API root; CelesTrak attribution on data views.

5. Acceptance Criteria

- All Must requirements demonstrably implemented; QA report (KSL-QA-001) shows tests passing.
- `uvicorn app.main:app` + `npm run dev` yields a working dashboard on localhost with Demo Mode data and no console errors.
- Unit tests green: Pc reference cases, CDM parser fixtures, screening on synthetic two-body setup, API contract tests.
- Documents delivered: SRS, HLD, LLD (PDF) + interactive pipeline diagram (HTML) + QA report.